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<!DOCTYPE html>
<html lang="ko">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0,
maximum-scale=1.0, user-scalable=no">
  <title>달려라 유우겐</title>
  <style>
    :root { --main: #FFB6C1; --point: #FF69B4; --bg: #FFD1DC; }
    body {
      margin: 0; overflow: hidden; background: var(--bg);
      font-family: 'Apple SD Gothic Neo', sans-serif;
      display: flex; justify-content: center; align-items: center; height: 100vh;
      touch-action: none;
    }
    #game-wrapper {
      position: relative; width: 100vw; height: 100vh;
      max-width: 800px; max-height: 450px;
      background: #fff; overflow: hidden;
      border: 5px solid white; border-radius: 20px;
      box-shadow: 0 20px 50px rgba(255, 105, 180, 0.3);
    }
    canvas { display: block; image-rendering: pixelated; width: 100%; height: 100%; }
    #ui {
      position: absolute; top: 20px; left: 20px; right: 20px;
      display: flex; justify-content: center; pointer-events: none; z-index: 10;
    }
    .stat-box {
      background: rgba(255, 255, 255, 0.9); border: 3px solid var(--point);
      padding: 10px 30px; border-radius: 50px; font-weight: bold;
      color: var(--point); font-size: 20px; box-shadow: 0 4px 0 var(--main);
    }
    .modal {
      position: absolute; top: 50%; left: 50%; transform: translate(-50%, -50%);
      background: white; padding: 40px; border-radius: 40px; border: 8px solid var(--main);
      text-align: center; box-shadow: 0 20px 60px rgba(0,0,0,0.2);
      z-index: 100; width: 300px;
    }
    h1 { color: var(--point); margin: 0 0 10px; font-size: 30px; }
    p { color: #777; margin-bottom: 25px; line-height: 1.6; }
    .btn {
      background: var(--point); color: white; border: none; padding: 15px 45px;
      font-size: 20px; border-radius: 50px; cursor: pointer; font-weight: bold;
      box-shadow: 0 6px 0 #d14789; transition: 0.1s;
    }
    .btn:active { transform: translateY(4px); box-shadow: 0 2px 0 #d14789; }
  </style>
</head>
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<body>

<div id="game-wrapper">
  <div id="ui">
    <div class="stat-box">SCORE: <span id="score">0</span></div>
  </div>

  <div id="startScreen" class="modal">
    <h1>달려라 유우겐</h1>
    <p>돌을 피해 점프하세요!<br><b>PC:</b> 스페이스바 / <b>모바일:</b> 터치</p>
    <button class="btn" onclick="startGame()">START</button>
  </div>

  <div id="gameOverScreen" class="modal" style="display: none;">
    <h1>GAME OVER</h1>
    <p>유우겐이 넘어졌어요!<br>최종 점수: <b id="finalScore" style="color:var(--point); font-size:24px;">0</b></p>
    <button class="btn" onclick="startGame()">RETRY</button>
  </div>

  <canvas id="gameCanvas"></canvas>
</div>

<script>
const canvas = document.getElementById("gameCanvas");
const ctx = canvas.getContext("2d");
const wrapper = document.getElementById("game-wrapper");

const img = {
  bg: new Image(), rock: new Image(), coin: new Image(),
  p1: new Image(), p2: new Image(), p3: new Image()
};
img.bg.src = '달려라배경.png';
img.rock.src = '달려라돌.png';
img.coin.src = '달려라코인.png';
img.p1.src = '달려라유우겐1.png';
img.p2.src = '달려라유우겐2.png';
img.p3.src = '달려라유우겐3.png';

let gameActive = false;
let score = 0;
let gameSpeed = 200; // 초당 이동 픽셀로 변경
let bgPos = 0;
let lastTime = 0;

const gravity = 1500; // 중력 가속도 상향 조정
let player = {
  x: 80, y: 0, w: 60, h: 60,

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    vy: 0, grounded: false,
    frame: 0, animTick: 0
  };

let obstacles = [];
let coins = [];

function resize() {
  canvas.width = wrapper.clientWidth;
  canvas.height = wrapper.clientHeight;
  player.y = canvas.height - 110;
}
window.onload = () => { resize(); draw(); };
window.onresize = resize;

const handleJump = () => {
  if (!gameActive) return;
  if (player.grounded) {
    player.vy = -550; // 점프 힘 조절
    player.grounded = false;
  }
};

window.addEventListener("keydown", (e) => { if(e.code === "Space") handleJump(); });
canvas.addEventListener("touchstart", (e) => { e.preventDefault(); handleJump(); });

function startGame() {
  gameActive = true;
  score = 0; gameSpeed = 200; obstacles = []; coins = [];
  bgPos = 0; lastTime = performance.now();
  document.getElementById("startScreen").style.display = "none";
  document.getElementById("gameOverScreen").style.display = "none";
  requestAnimationFrame(update);
}

function update(currentTime) {
  if (!gameActive) return;

  // 델타 타임 계산 (초 단위)
  const dt = (currentTime - lastTime) / 1000;
  lastTime = currentTime;

  // 프레임 드랍이나 탭 전환 시 과도한 점프 방지
  if (dt > 0.1) { requestAnimationFrame(update); return; }

  bgPos -= gameSpeed * 0.5 * dt;
  if (Math.abs(bgPos) >= canvas.width) bgPos = 0;

  player.vy += gravity * dt;

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player.y += player.vy * dt;

const groundY = canvas.height - 110;
if (player.y > groundY) {
    player.y = groundY;
    player.vy = 0;
    player.grounded = true;
}

player.animTick += dt;
if (player.animTick > 0.12) { // 0.12초마다 프레임 변경
    player.frame = (player.frame + 1) % 3;
    player.animTick = 0;
}

// 장애물 생성
if (Math.random() < 1.2 * dt) {
    if (obstacles.length === 0 || (canvas.width - obstacles[obstacles.length-1].x > 300))
    {
        obstacles.push({ x: canvas.width, y: canvas.height - 75, w: 30, h: 30, passed:
false });
    }
}

obstacles.forEach((obs, i) => {
    obs.x -= gameSpeed * dt;
    if (obs.x + obs.w < player.x && !obs.passed) {
        score += 10;
        obs.passed = true;
    }
}

// 히트박스 판정
let pLeft = player.x + 22, pRight = player.x + player.w - 22;
let pTop = player.y + 12, pBottom = player.y + player.h - 5;
let oLeft = obs.x + 10, oRight = obs.x + obs.w - 10;
let oTop = obs.y + 5, oBottom = obs.y + obs.h;

if (pLeft < oRight && pRight > oLeft && pTop < oBottom && pBottom > oTop) {
    gameOver();
}
if (obs.x < -50) obstacles.splice(i, 1);
});

// 코인
if (Math.random() < 0.8 * dt) {
    coins.push({ x: canvas.width, y: canvas.height - 210 - Math.random()*40, w: 30, h:
30 });
}

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    coins.forEach((c, i) => {
        c.x -= gameSpeed * dt;
        if (player.x < c.x + c.w && player.x + player.w > c.x &&
            player.y < c.y + c.h && player.y + player.h > c.y) {
            score += 5;
            coins.splice(i, 1);
        }
        if (c.x < -50) coins.splice(i, 1);
    });

    if (score < 300) {
        gameSpeed += 5 * dt; // 점진적 속도 증가 (300점 미만일 때만)
    }
    document.getElementById("score").innerText = score;

    draw();
    requestAnimationFrame(update);
}

function draw() {
    ctx.clearRect(0, 0, canvas.width, canvas.height);
    ctx.drawImage(img.bg, Math.floor(bgPos), 0, canvas.width + 2, canvas.height);
    ctx.drawImage(img.bg, Math.floor(bgPos + canvas.width), 0, canvas.width + 2,
canvas.height);

    let plmg = player.grounded ? [img.p1, img.p2, img.p3][player.frame] : img.p2;
    ctx.drawImage(plmg, player.x, player.y, player.w, player.h);

    obstacles.forEach(obs => ctx.drawImage(img.rock, obs.x, obs.y, obs.w, obs.h));
    coins.forEach(c => ctx.drawImage(img.coin, c.x, c.y, c.w, c.h));
}

function gameOver() {
    gameActive = false;
    document.getElementById("gameOverScreen").style.display = "block";
    document.getElementById("finalScore").innerText = score;
}
</script>
</body>
</html>

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